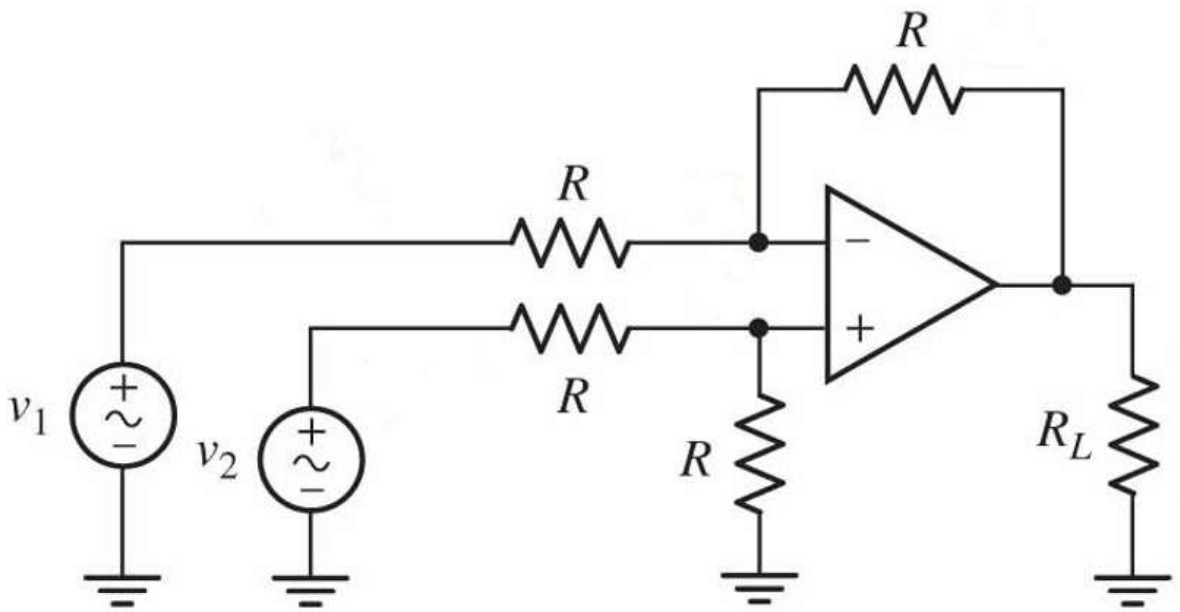
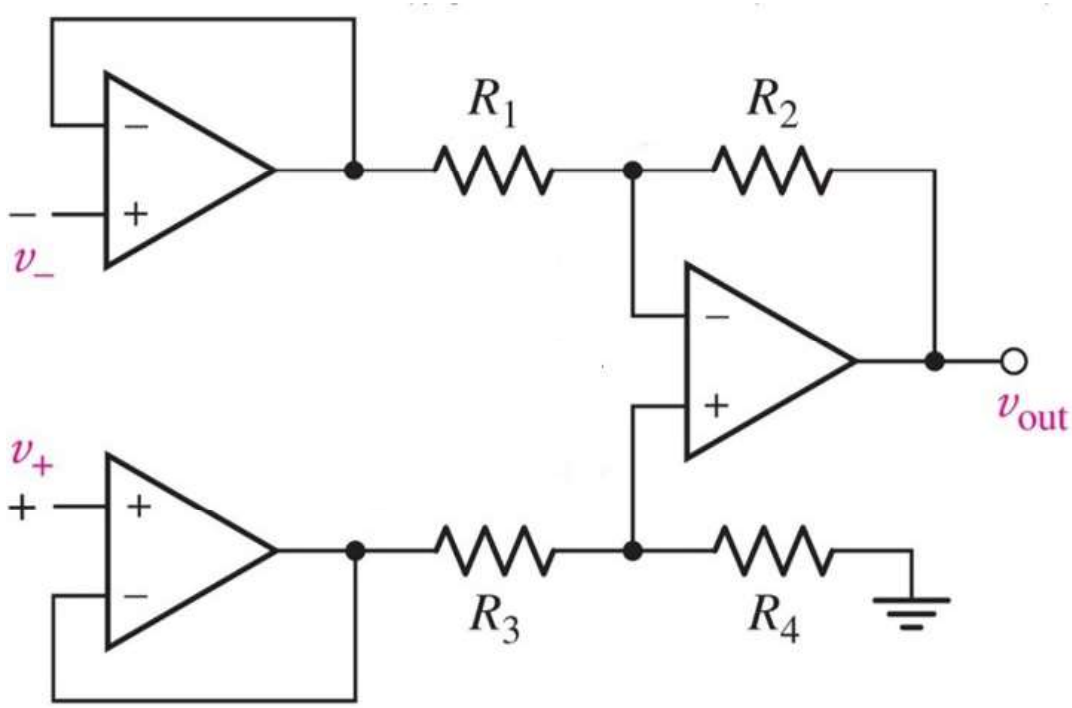
Tutorial question 1

Derive an expression for v_{out} in terms of v_1 and v_2 for the circuit shown below, which is known as a difference amplifier.



Tutorial question 2

If $R_4/R_3 = R_2/R_1 = K$, derive an expression for v_{out} in terms of v_+ and v_- for the circuit shown below, which is known as the instrumentation amplifier.



Tutorial question 3

Design a circuit using two ideal op amps to produce the following output relationship:

$$v_{out} = 2v_1 + 3v_2$$

Assume both op amps operate in the linear region. Use an inverting summing amplifier followed by an inverting amplifier. All resistors must lie between 10 kΩ and 100 kΩ.

(a) Draw your circuit and label all resistor values and the intermediate output v_x.

(b) Derive expressions for both stage outputs to verify your design.

(c) Calculate both stage outputs when $v_1 = 0.5\ \text{V}$ and $v_2 = -0.2\ \text{V}$.

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